

IssueFlow Docker Environment

IssueFlow — Docker Environment Documentation

Overview

The IssueFlow application runs as a multi-service Docker Compose stack consisting of four services: a PostgreSQL database, a backend API, a frontend web server, and a self-hosted GitLab instance.

Services

1. db — PostgreSQL Database

Property	Value
Image	postgres:16-alpine
Restart policy	unless-stopped
Host port	5432
Container port	5432
Persistent volume	postgres_data → /var/lib/postgresql/data

Environment variables (with defaults):

Variable	Default
POSTGRES_DB	issueflow
POSTGRES_USER	issueflow
POSTGRES_PASSWORD	issueflow

Health check: Runs pg_isready every 5 seconds (10 retries, 5 s timeout). Other services that depend on this will not start until the health check passes.

2. backend — API Server

Property	Value
Build context	./backend
Restart policy	unless-stopped
Host port	4000 (configurable)
Container port	4000 (configurable)
Persistent volume	uploads_data → /app/uploads
Depends on	db (healthy)

Environment variables (with defaults):

Variable	Default
PORT	4000
DATABASE_URL	postgres://issueflow:issueflow@db:5432/issueflow
CORS_ORIGIN	http://localhost:5173

The backend waits for the db service to report healthy before it starts.

3. frontend — Web Application

Property	Value
Build context	./frontend
Restart policy	unless-stopped
Host port	5174
Container port	80
Depends on	backend

Build arguments:

Argument	Value
VITE_API_URL	<i>(empty — resolved at runtime via relative paths)</i>

The frontend is built with an empty VITE_API_URL so that API calls are made relative to the host, allowing the app to work behind a reverse proxy without rebuilding. The `host.docker.internal` extra host entry is added to allow the container to reach services on the Docker host machine.

4. gitlab — GitLab Community Edition

Property	Value
Image	gitlab/gitlab-ce:latest
Restart policy	unless-stopped
Hostname	gitlab.local (configurable)
shm_size	256 MB

Ports:

Purpose	Host port	Container port
HTTP	8929	8929
HTTPS	4430	443
SSH	2224	22

Persistent volumes:

Volume	Mount path
gitlab_config	/etc/gitlab
gitlab_logs	/var/log/gitlab
gitlab_data	/var/opt/gitlab

Omnibus configuration (with defaults):

Setting	Default
external_url	http://localhost:8929
gitlab_shell_ssh_port	2224

Volumes

All named volumes are managed by Docker and persist across container restarts.

Volume	Used by	Purpose
postgres_data	db	PostgreSQL database files
uploads_data	backend	User-uploaded files
gitlab_config	gitlab	GitLab configuration
gitlab_logs	gitlab	GitLab log files

Volume	Used by	Purpose
gitlab_data	gitlab	GitLab application data

Service Dependency Graph

```
gitlab  (independent)
|
frontend → backend → db (healthcheck)
```

Environment Variable Overrides

All configurable values can be set in a `.env` file in the project root:

```
# Database
POSTGRES_DB=issueflow
POSTGRES_USER=issueflow
POSTGRES_PASSWORD=issueflow

# Backend
BACKEND_PORT=4000
BACKEND_DATABASE_URL=postgres://issueflow:issueflow@db:5432/issueflow
BACKEND_CORS_ORIGIN=http://localhost:5173

# GitLab
GITLAB_HOSTNAME=gitlab.local
GITLAB_EXTERNAL_URL=http://localhost:8929
GITLAB_HTTP_PORT=8929
GITLAB_HTTPS_PORT=4430
GITLAB_SSH_PORT=2224
```

Quick Start

Start all services in the background

```
docker compose up -d
```

View logs

```
docker compose logs -f
```

Stop all services

```
docker compose down
```

Stop and remove volumes (destructive)

`docker compose down -v`

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